

SAMPLE IDENTIFICATION	
Code	Faro 21
Pit	1-GU-2-PA
Depth	1,740.10 m
Coordinates (well)	Lat. 1°20'54.75" S Long. 51°25'32.84" W
Formation / Basin	Lighthouse / Amazon Basin
Age	Tournaisian – Viséan (Lower Carboniferous)

GENERAL DESCRIPTION OF THE BLADE
Clastic textured lamina, grain-supported framework with contribution of fine intergranular material. It differs from Faro 12 by the higher proportion of dark grains (dark gray to black under N+), suggestive of lithic fragments and/or intensely altered feldspar, giving a more immature compositional aspect. Brownish/orange spots and films are evident in the interstices and intergranular contacts, indicating significant ferruginous cementation. A slightly higher concentration of fine intergranular material (possibly clayey/ferruginous matrix) is perceptible in sectors C and SE. There are no visible fractures. Open primary porosity is not evident (interstices filled with cement/fine material).

Estimated modal composition (% vol)	
Monocrystalline quartz (Qz mono)	~55–60%
Polycrystalline quartz	~5–8%
Feldspar (K-feldspar + plagioclase)	~8–12%
Lithic fragments	~8–12%
Micas (muscovite ± biotite)	~2–3%
Opaque minerals	~3–5%
Cement (ferruginous + siliceous)	~5–8%
Clayey/ferruginous matrix	~3–5%
Visible porosity	<1%

Composition:

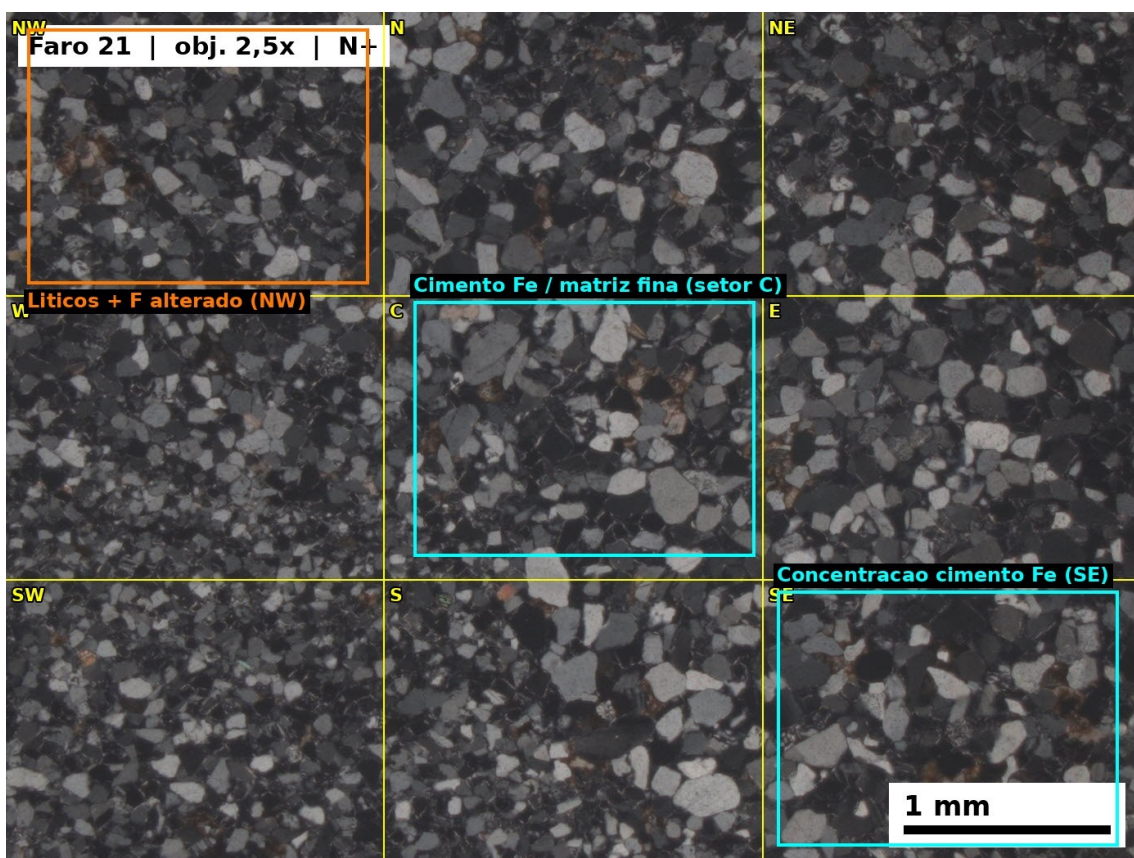
Lower compositional maturity of the assemblage (F+L ~16–24%). Expressive lithic and feldspathic fraction : sublitharenite to lithoarenite (Folk, 1974). Higher proportion of dark grains may reflect distributary channel facies with less reworking , preserving compositionally immature grains. Micaceous character compatible with micaceous sandstones of the Faro Formation. Abundant ferruginous cementation consistent with ferruginous sandstones. Variable composition, ranging from quartzose to feldspathic and orthoquartzitic .

Microscopic description	
Texture	Terrigenous clastic, grain-supported with a

	contribution of fine intergranular material, homogeneous.
Particle size	Fine to medium sand (~0.12–0.40 mm). Mode: fine to medium sand (~0.18–0.25 mm).
Particle size selection	Moderately to moderately poorly selected.
Packaging and contacts	Dense. Long (~40%), concave-convex (~35%), sutured (~15%), punctate (~10%).
Cement / diagenesis	Abundant ferruginous cementation (hematite/goethite), filling interstices and forming grain coatings (sectors C and SE). Subordinate siliceous cementation (overgrowth) . Qz) by sutured contacts. Sequence: mechanical compaction → infiltration of clays + Fe-oxides → chemical compaction + overgrowth Qz → delayed oxidation.
Porosity	No evident open primary porosity. Interstices occupied by ferruginous cement and clayey material. Possible microporosity in clay minerals resulting from feldspar alteration.
Maturity	Textural : underripe (moderate to poor sorting, subangular grains , matrix >3%). Compositional: submature to immature (low Qt/F+L). Lower maturity suggests less reworking , consistent with a proximal fluvio -deltaic facies.

Sectoral analysis of grains (3x3 grid)					
Sector	Mineral	Size (mm)	Sphericity	Rounding	Form
NW	Qz mono	0.20	Average	Subangular	Equidimensional
NW	Lithic	0.30	Low	Subangular	elongated
NW	Opaque	0.10	High	Rounded	Equidimensional
N	Qz mono	0.25	High	Subrounded	Equidimensional
N	Feldspar	0.20	Average	Subangular	Prismatic
N	Lithic	0.35	Low	Angular	elongated
NE	Qz mono	0.15	Average	Subrounded	Equidimensional
NE	Qz poli	0.25	Low	Subangular	elongated
W	Qz mono	0.22	High	Subrounded	Equidimensional
W	Lithic	0.28	Low	Subangular	Discoidal
W	Feldspar	0.18	Average	Subangular	Prismatic
W	Qz mono	0.20	Average	Subangular	Equidimensional
W	Opaque	0.12	High	Rounded	Equidimensional
AND	Qz mono	0.30	High	Subrounded	Equidimensional

AND	Lithic	0.22	Low	Angular	elongated
SW	Qz mono	0.18	Average	Subangular	Equidimensional
SW	Feldspar	0.25	Average	Subangular	Prismatic
S	Qz mono	0.15	High	Subrounded	Equidimensional
S	Qz mono	0.20	Average	Subrounded	Equidimensional
SE	Qz mono	0.22	High	Subrounded	Equidimensional
SE	Lithic	0.35	Low	Angular	elongated
SE	Qz poli	0.20	Low	Subangular	elongated



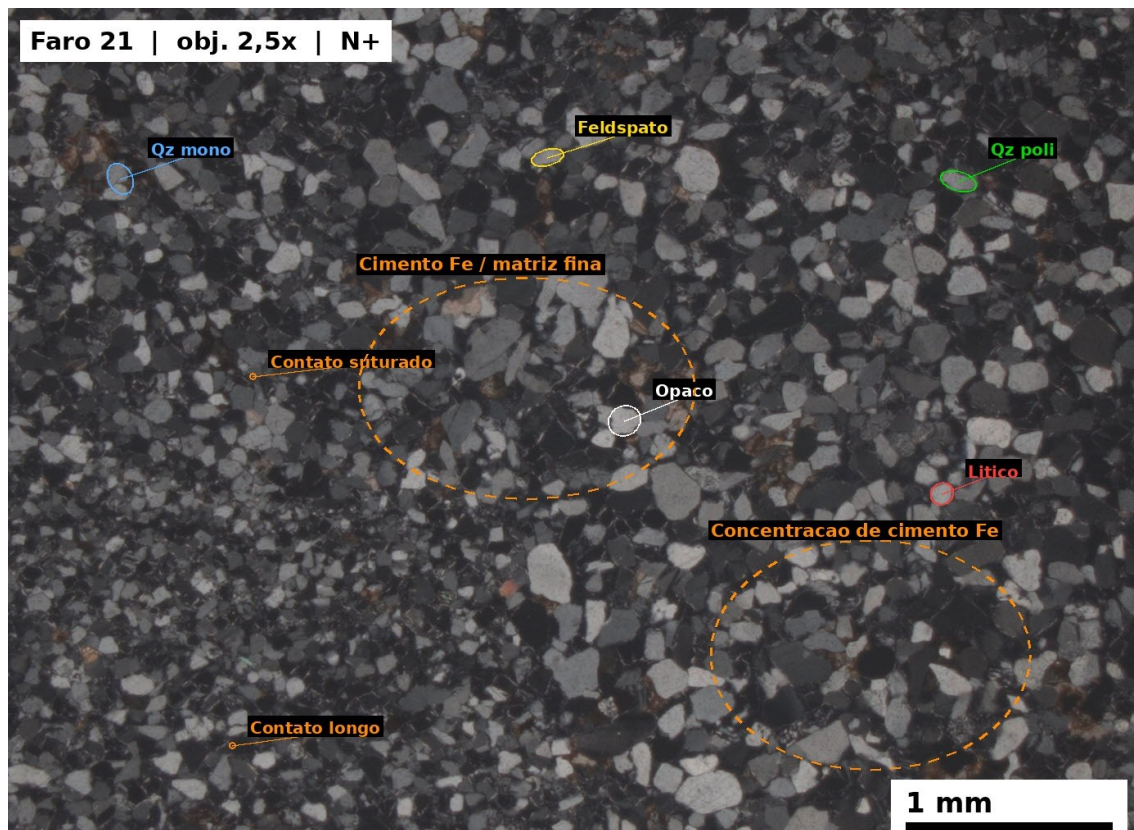
Grain population:

Dominant population: Qz mono, medium-high sphericity, subangular – subrounded , equidimensional. Significant subordinate populations: (1) lithics, low sphericity, angular– subangular , elongated; (2) altered feldspar, medium sphericity, subangular , prismatic.

CLASSIFICATION

Sublithoarenite to fine to medium-grained lithoarenite , moderately sorted, submature .

PHOTOMICROGRAPHY



Photomicrograph (N+, 2.5× objective) of fine- to medium- grained , grain-supported sublitharenite with a fine intergranular matrix. Higher proportion of lithic fragments (L) and altered feldspar (F) compared to other samples in the assemblage, associated with dominant monocrystalline quartz (Qz mono) and subordinate polycrystalline quartz (Qz poly). Abundant ferruginous cementation filling interstices and contacts, accompanied by an increase in sutured contacts related to the larger unstable fraction of the framework. Scale bar ≈ 1 mm.